

Probability Spaces - (Ω, F, P) defines a random experiment.

Sample Space (Ω) - the set of all possible outcomes of the experiment.

- Ex: Pick 1 card from deck
- {A spades, A clubs, K diamonds, ...}

Event Space (F) - events in your experiment, subsets of Ω

- $\Omega \in F$, the whole sample space is an event
- Closed under complement (if "all red cards" is an event, so is "all non red cards".)
- Closed under countable unions (if $A_1, A_2, \dots \in F$, then $\bigcup_i A_i \in F$)
- Ex: all red cards, all black cards, $\emptyset, F$

Probability Function (P) - assigns a probability to each event.

$$- P(\emptyset) = 0 \quad P(E) \geq 0 \quad \forall E \in F \text{ (Nonnegativity)}$$

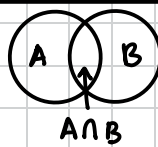
$$- P(\Omega) = 1$$

- Countable additivity (probs add up for disjoint events)

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i) \text{ for disjoint events } E_i$$

Bayes Rule

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$



Total Probability

If $B_1, \dots, B_n$ are disjoint & partition the sample space, $P(A) = \sum_i P(A \cap B_i) = \sum P(A|B_i)P(B_i)$

Independence

Knowing one event occurred doesn't change the probability of another event.

$$P(A|B) = P(A) \iff P(A \cap B) = P(A)P(B)$$

Inclusion Exclusion

When finding $P(A \cup B)$, adding their individual probabilities double counts the shared outcome.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \left(\sum_{1 \leq i_1 < \dots < i_k \leq n} P(A_{i_1} \cap \dots \cap A_{i_k}) \right) \text{ and}$$

$$P\left(\bigcap_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \left(\sum_{1 \leq i_1 < \dots < i_k \leq n} P(A_{i_1} \cap \dots \cap A_{i_k}) \right)$$

$$\begin{aligned} P(A \cup B) &= P(A \cup (B|A)) \\ &= P(A) + P(B|A) \\ &= P(A) + P(B|A) + P(A \cap B) - P(A \cap B) \\ &= P(A) + P(B|A) \cup (A \cap B) - P(A \cap B) \\ &= P(A) + P(B) - P(A \cap B) \end{aligned}$$

Union Bound - extension of inclusion exclusion

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i)$$

Random Variables - function from sample space (Ω) to a subset of the real numbers $(\mathbb{R})$

Ex: 2 coin flips, # of heads - $\Omega = \{HH, HT, TH, TT\}$

$$\text{Let } X = \# \text{ of heads: } X: \Omega \rightarrow \{0, 1, 2\} \quad \begin{array}{ll} X(HH) = 2 & X(HT) = 1 \\ X(TH) = 1 & X(TT) = 0 \end{array}$$

Discrete RVs have probability mass functions:

$$P_X(x) = P(\{X=x\}) = P(\{\omega \in \Omega \mid X(\omega) = x\})$$

Calculate by looping over every outcome ω in Ω , keep only the ones where $X(\omega) = x$, and add their probabilities.

1 $\{x\}$ is an indicator function which returns 1 if x is true, else 0.

$$\text{Ex: } P_X(0) = P(\{TT\}) = 1/4, \quad P_X(1) = P(\{HT, TH\}) = 1/2, \quad P_X(2) = P(\{HH\}) = 1/4$$

Expectation - Center of mass of a distribution.

$$E[X] = \sum_{a \in A} a \cdot P[X=a] = \sum_x x \cdot P_X(x)$$

Linearity of Expectation: $E[X+Y] = E[X] + E[Y]$

$$E[cX] = cE[X], \quad \forall c \in \mathbb{R}$$

Law of the Unconscious Statistician (LOTUS):

$$E[f(X)] = \sum_x f(x) P(X=x) \quad \text{For example, } E[X^2] = \sum_x x^2 P[X=x]$$

Conditional Expectation: $E[X|Y=y] = \sum_x x \cdot P_{X|Y}(x|y)$

$$\text{Total Expectation Theorem: } E[X] = \sum_y P_Y(y) \cdot E[X|Y=y]$$

Variance - Average squared distance of a random variable from its mean.

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) \text{ if independent.}$$

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

$$\text{Var}(cX) = c^2 \text{Var}(X)$$

$$\text{Var}(X+b) = \text{Var}(X)$$

Bernoulli Distribution

X is a Bernoulli random variable with parameter p . $X \in \{0, 1\}$. $P(X=1) = p$, $P(X=0) = 1-p$.

Let I_A be the Bernoulli variable for event A w/ parameter $P(A)$.

$$E[I_A] = P(A) = p$$

$$\text{Var}(I_A) = p(1-p)$$

Binomial Distribution

Models the number of successes in n independent identical trials, where each trial succeeds with probability p . For example, flipping a coin multiple times. Converges to Poisson as $n \rightarrow \infty$

Let $X_i = \begin{cases} 1 & \text{if flip } i \text{ is H} \\ 0 & \text{otherwise} \end{cases}$

$X_i \sim \text{Bernoulli}(p)$ X_i is Bernoulli RV with probability p

$$\text{Let } X = \sum_{i=1}^n X_i \quad P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$X \sim \text{Bin}(n, p)$ X is Binomial RV with n trials and probability p .

$$E[X] = \sum_{i=1}^n E[X_i] = \sum_{i=1}^n p = np \quad \text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) = \sum_{i=1}^n p(1-p) = np(1-p)$$

Geometric Distribution

Models how many trials are needed until the first success. Let X = # of flip until first H.

$X \sim \text{Geometric}(p)$ X is a geometric RV with probability p .

$$P(X=x) = (1-p)^{x-1} p \quad E[X] = \frac{1}{p} \quad \text{Var}(X) = \frac{1-p}{p^2}$$

Memoryless property: If m trials have already failed, the remaining waiting time is distributed the same as if you are starting over. Essentially, the coin doesn't remember its past flips.

$$P(X > m+n | X > m) = P(X > n)$$

Poisson Distribution

Models how many times an event happens in a fixed interval.

λ = average number of events per interval

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad E[X] = \lambda \quad \text{Var}(X) = \lambda$$

Uniform Distribution

A random variable is uniform on an interval if every value of the RV within the interval is equally likely

$X \sim \text{Uniform}(a, b)$, where a, b is the interval

$$P(X=x) = \frac{1}{b-a+1} \quad E[X] = \frac{a+b}{2} \quad \text{Var}(X) = \frac{(b-a+1)^2 - 1}{12}$$

Indicator Tricks

Useful to decompose RVs into sums of Bernoulli RVs

Ex: Let X = # of H in n coin flips

Let $RV = I_i = \begin{cases} 1 & \text{Flip } i \text{ is heads} \\ 0 & \text{otherwise} \end{cases}$, $X = \sum_{i=1}^n I_i$

$$\text{Then, } E[X] = E\left[\sum_{i=1}^n I_i\right] = \sum_{i=1}^n E[I_i] = \sum_{i=1}^n \frac{1}{2} = \frac{n}{2}$$

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^n I_i\right) = \sum_{i=1}^n \text{Var}(I_i) = \sum_{i=1}^n \frac{1}{4} = \frac{n}{4}$$

Calculus

Fundamental Theorem: $\frac{d}{dx} \int_a(x)^{b(x)} f(t) dt = f(b(x))b'(x) - f(a(x))a'(x)$

$$\int_a^b f(t) dt = F(b) - F(a), \text{ where } F \text{ is antiderivative of } f$$

Chain Rule: $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$

Discrete vs. Continuous

$$\begin{array}{ll} \text{Discrete} & \text{Continuous} \\ \text{Expectation } E[X] = \sum_{a \in A} a \cdot P[X=a] & E[X] = \int_{-\infty}^{\infty} a \cdot f_X(a) da \end{array}$$

$$\text{LOTUS } E[g(X)] = \sum_x g(x) P(X=x) \quad E[g(X)] = \int_{-\infty}^{\infty} g(t) f_X(t) dt$$

$$\text{Total Prob } P[B] = \sum_{i=1}^n P[B|A_i]P[A_i] \quad P[B] = \int_{-\infty}^{\infty} P[B|A_i] f_X(t) dt$$

$$\text{Variance } \text{Var}(X) = E[X^2] - (E[X])^2 \quad \text{Var}(X) = \int_{-\infty}^{\infty} x^2 f(x) dx - \left(\int_{-\infty}^{\infty} x f(x) dx\right)^2$$

Law of Total Variance: $\text{Var}(Y) = E[\text{Var}(Y|X)] + \text{Var}(E[Y|X])$

Continuous Random Variable

Takes uncountably many values (any real number in an interval)
 $P(X=a) = 0$ because a single point has no width. Instead of $P(X=a)$, $P(a \leq X \leq b)$. What is the probability that a RV lies in some interval?

Probability Density Function

The PDF is the actual curve of the distribution. Probability is the area under the curve.

Properties:

- ① Non-negativity: $f_X(x) \geq 0$ for all x . $E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x) dx = \int_0^{\infty} P(X \geq x) dx$
- ② Interval Probability: $P(a \leq X \leq b) = \int_a^b f_X(x) dx$
- ③ Normalization: $\int_{-\infty}^{\infty} f_X(x) dx = 1$ Total probability sums to 1

Cumulative Distribution Function

$F_X(a) = P(X \leq a)$. Gives the probability that X is to the left of a . $P(a \leq X \leq b) = P(a < X < b)$

$$F_X(a) = \int_{-\infty}^a f_X(x) dx \quad \frac{d}{dx} F_X(x) = f_X(x)$$

Properties:

- ① Non decreasing: $F_X(x)$ never goes down, only add probability as you go right. $\rightarrow$ $\rightarrow$ and $\rightarrow$ refer to the bounds of the domain of X . So if X is only over c to d , use c and d instead of $-\infty$ and ∞ .
- ② Limits: $\lim_{x \rightarrow -\infty} F_X(x) = 0$ $\lim_{x \rightarrow \infty} F_X(x) = 1$

Uniform Distribution

A random variable is uniform on an interval if every value of the RV within the interval is equally likely

$X \sim \text{Uniform}(a, b)$, where a, b is the interval

$$\text{PDF: } f_X(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases} \quad \text{CDF: } F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}$$

$$P(c \leq x \leq d) = \frac{d-c}{b-a} = \frac{\text{interval of probability}}{\text{interval of distribution}}$$

$$E[X] = \frac{a+b}{2} \quad \text{Var}(X) = \frac{(b-a)^2}{12}$$

Exponential Distributions

Models waiting time until an event happens

$X \sim \text{Exp}(\lambda) \rightarrow X$ = time until first success, λ = rate at which successes happen

$$\text{PDF: } f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{CDF: } F_X(t) = P(X \leq t) = 1 - e^{-\lambda t}$$

$$P(a \leq x \leq b) = \int_a^b \lambda e^{-\lambda x} dx \quad \text{by } t$$

Survival function: Probability that object survives past + $S_X(t) = P(X > t) = 1 - F_X(t) = e^{-\lambda t}$

$$E[X] = \frac{1}{\lambda} \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

Memoryless property: If m time has already past, the additional waiting time does not depend on the past.

$$\begin{aligned} P(X > m+n | X > m) &= P(X > n) \\ P(X > a+b | X > a) &= \frac{P(X > a+b, X > a)}{P(X > a)} = \frac{P(X > a+b)}{P(X > a)} = \frac{1 - F_X(a+b)}{1 - F_X(a)} \\ &= \frac{e^{-\lambda(a+b)}}{e^{-\lambda a}} = e^{-\lambda b} = P(X > b) \end{aligned}$$

Normal Distribution

$X \sim N(\mu, \sigma^2)$ means X is normally distributed with mean μ and variance σ^2

$$\text{PDF: } f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$$

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{x-\mu}{\sigma}} e^{-\frac{z^2}{2}} dz$$

$$E[X] = \mu \quad \text{Var}(X) = \sigma^2 \quad \begin{array}{l} X \sim N(\mu, \sigma^2) \\ Z = \frac{X - \mu}{\sigma} \end{array}$$

Sum of Independent Normals: $Z \sim N(0, 1)$
 $X \sim N(0, 1)$ and $Y \sim N(0, 1)$
 $Z = aX + bY \sim N(0, a^2 + b^2)$
 $P(X \leq x) = P(Z \leq \frac{x-\mu}{\sigma})$

Joint Densities

$$P((x, y) \in A) = \iint_A f_{X,Y}(x, y) dx dy$$

$$\text{Normalization condition: } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$$

$$\text{Marginal Density: } f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

Law of Total Expectation: $E[X] = E[E[X|Y]]$

Ex: $Z \sim N(\mu, \sigma^2)$ & $A \sim \text{Uni}(0, Z^2)$. What is $E[A]$?

$$E[Z^2] = \text{Var}(Z) + (E[Z])^2 = \sigma^2 + \mu^2$$

$$\begin{aligned} E[A] &= E[E[A|Z]] = E\left[\frac{0+Z^2}{2}\right] = E\left[\frac{Z^2}{2}\right] = \frac{1}{2} E[Z^2] = \frac{1}{2} (\sigma^2 + \mu^2) \\ E[E[X|Y]] &= \sum_i E[X|Y=y_i] P(Y=y_i) = \sum_i \left(\sum_j x_j P(X=x_j | Y=y_i)\right) P(Y=y_i) \\ &= \sum_i x_j \sum_j P(X=x_j | Y=y_i) P(Y=y_i) = \sum_j x_j P(X=x_j) = E[X] \end{aligned}$$

Poisson Distribution

Models how many times an event happens in a fixed interval.

λ = average number of events per interval

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad E[X] = \lambda \quad \text{Var}(X) = \lambda$$

Convolutions

Let $Z = X + Y$ for independent X & Y

Discrete Case: $P(Z=z) = \sum_x P(Z=z | X=x) P(X=x)$
 $= \sum_x P(X+Y=z | X=x) P(X=x)$
 $= \sum_x P(Y=z-x | X=x) P(X=x)$

Convolution Formula: $P(Z=z) = \sum_x P(Y=z-x) P(X=x)$

Continuous case: $f_z(z) = \int_{-\infty}^{\infty} f_x(x) f_y(z-x) dx$

Covariance: Measures the linear association between two random variables

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = \frac{P(X \wedge Y) - P(X)P(Y)}{P(X)P(Y)}$$

If X and Y are independent, the $\text{Cov}(X, Y) = 0$ b/c $E[XY] = E[X]E[Y]$

① $\text{Cov}(X, Y) = \text{Cov}(Y, X)$ ② $\text{Cov}(aX + bY, cZ) = ac \text{Cov}(X, Z) + bc \text{Cov}(Y, Z)$
③ $\text{Cov}(X, X) = \text{Var}(X)$

Moment Generating Functions (MGF)

$$M_X(t) = E[e^{tX}]$$

MGF - function of t that encodes all the information about a distribution in one object.

Use 1: Extract Moments

$$E[X^k] = \left. \frac{d^k}{dt^k} M_X(t) \right|_{t=0}$$
 The k th moment of X is the k th derivative of the MGF evaluated at $t=0$.

Use 2: Compute Sums

For independent X and Y : $M_{X+Y}(s) = E[e^{s(X+Y)}] = E[e^{sX}]E[e^{sY}] = M_X(s) \cdot M_Y(s)$

Use 3: Affine Transforms

For $aX + b$: $M_{aX+b}(s) = e^{bs} E[e^{asX}] = e^{bs} M_X(as)$
P.E. $E[e^{aX+b}] = E[e^{asX} \cdot e^{bs}] = e^{bs} E[e^{asX}] = e^{bs} M_X(as)$

RV	MGF	RV	MGF
Bernoulli	$1 - p + pe^t$	Normal	$e^{t\mu + \sigma^2 t^2 / 2}$
Binomial	$(1 - p + pe^t)^n$	Exponential	$\frac{1}{1 - \frac{t}{\lambda}}$
Geometric	$\frac{p}{1 - (1-p)e^t}$	Poisson	$e^{\lambda(e^t - 1)}$

Bounds

Markov: For a nonnegative RV: $P(X \geq a) \leq \frac{E[X]}{a}$

(Chebyshev: For $b \geq 0$: $P(|X - E[X]| \geq b) \leq \frac{\text{Var}(X)}{b^2}$

(Chernoff: $P(X \geq a) = P(e^{tX} \geq e^{ta}) = \frac{E[e^{tX}]}{e^{ta}} = \frac{M_X(t)}{e^{ta}}$

Convergence in Probability

For every $\epsilon > 0$, $\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0$ The probability that X_n differs from X by more than ϵ shrinks to 0 as $n \rightarrow \infty$.

Weak Law of Large Numbers - Sample mean converges to expected value.

$X_1, X_2, \dots, X_n$ are i.i.d. with mean μ and a finite variance.

$$P\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| \geq \epsilon\right) \rightarrow 0 \text{ as } n \rightarrow \infty$$

As $n \rightarrow \infty$, the probability that the mean of $X_n - \mu$ is $\geq \epsilon$ is 0.

Strong Law of Large Numbers

$P(\lim_{n \rightarrow \infty} \bar{X}_n = \mu) = 1$. The probability of the average mean of all X_n 's equaling μ as $n \rightarrow \infty$ is 1.

Central Limit Theorem

When n is large, the distribution of the sample mean or sample sum of identical distribution RVs looks Normal, regardless of the original distribution.

$X_1, X_2, X_3, \dots, X_n$: Independent and identically distributed (i.i.d.) RVs

Each X_i has mean, μ , and standard deviation, σ .

Sample Sum: $S_n = \sum_{i=1}^n X_i$ $E[S_n] = n\mu$ $SD(S_n) = \sqrt{n}\sigma$

$S_n \approx N(n\mu, n\sigma^2)$ for large n

Sample Mean: $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ $E[\bar{X}_n] = \mu$ $SD(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$

$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$ for large n $Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$

You can define $Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}$, then $E(Z_n) = 0$, $\text{Var}(Z_n) = 1$.

$\lim_{n \rightarrow \infty} P(Z_n \leq x) = \Phi(x)$ $\forall x$ when $\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$

Entropy

$$H(X) = E[-\log_2 p(X)] = E\left[\log_2 \frac{1}{p(X)}\right] = \sum_{x \in X} p(x) \cdot \log_2 \left(\frac{1}{p(x)}\right) \text{ bits}$$

- average amount of uncertainty/surprise in X .

$-\log_2 \frac{1}{p(x)}$ measures how surprising a single outcome x is.

- If high $p(x)$, low $\log_2 \frac{1}{p(x)} \rightarrow$ high prob, not surprising

- If low $p(x)$, high $\log_2 \frac{1}{p(x)} \rightarrow$ low prob, surprising

- High entropy $\Rightarrow$ more uncertainty $\Rightarrow$ learning the outcome tells you more info

Asymptotic Equipartition Property (AEP)

If $X_1, X_2, \dots, X_n$ are i.i.d. $\sim p(x)$: $-\log_2 p(X_1, X_2, \dots, X_n) \xrightarrow{p} H(X)$

P.E. Since X_i are independent:

$$-\log_2 p(X_1, \dots, X_n) = -\sum_{i=1}^n \log_2 p(X_i)$$

Divide by $n = \frac{1}{n} \sum_{i=1}^n -\log_2 p(X_i)$, which is the sample mean of $-\log_2 p(X_i)$

By WLLN, sample means converges to expected value so

$$\frac{1}{n} \sum_{i=1}^n -\log_2 p(X_i) \xrightarrow{p} E[-\log_2 p(X)] = H(X)$$

Typical Sequences

A sequence x^n is typical if $|\frac{1}{n} \log_2 p(x^n) - H(X)| \leq \epsilon$ OR $2^{-n(H(X) + \epsilon)} \leq p(x^n) \leq 2^{-n(H(X) - \epsilon)}$

About 2^{nH} typical sequences, each w/ probability about 2^{-nH} .

Mutual Information

- average amount of info X provides about Y and vice versa (symmetric)

$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) = H(X) + H(Y) - H(X, Y)$$

If X, Y independent, $I(X; Y) = 0$.

Joint Entropy - How surprising is X given Y ?

$$H(X, Y) = -\sum_{x,y} p(x,y) \cdot \log(p(x,y))$$

$$= -\sum_{x,y} p(x,y) \cdot \log(p(y) \cdot p(x|y))$$

$$= -\sum_{x,y} p(x,y) \cdot \log(p(y)) - \sum_{x,y} p(x,y) \cdot \log(p(x|y))$$

$$= -\sum_y p(y) \cdot \log(p(y)) - \sum_y p(y) \sum_x p(x|y) \log(p(x|y))$$

$$H(X, Y) = H(Y) + H(X|Y)$$

Channels

Sender chooses an input X with distribution $p(x)$. Sends it thru a noisy channel and the receiver receives Y . Channel behaves according to $p(y|x)$, probability of receiving y given you sent x . Sender controls $p(x)$ but not the channel $p(y|x)$.

Types of channels:

Noiseless: Whatever you send arrives perfectly, no distortion $p(y|x) = \delta_{xy}$

Noisy: Signal is corrupted

BSC - bits flip w/ probability p , ($= 1 - I(p)$)

BEC - bits are erased w/ probability p , ($= 1 - p$)

Gaussian - random noise $Y = X + Z$, where $Z \sim N(0, \sigma^2)$

Fundamental Question: How much info can be reliably sent through this channel? Same as how of X can the receiver Y retain? This is mutual information $I(X; Y)$. Maximize this over all possible input distributions $p(x)$.

($= \max_{p(x)} I(X; Y)$), C is the channel capacity, the max rate at which info can be transmitted reliably, measured in bits per channel use. Reliable iff $m/n \leq C$. Max $m = nC$

KL Divergence

$D(p||q) = \sum_x p(x) \cdot \log \frac{p(x)}{q(x)}$. This measures how different two distributions p and q are. $D(p||q)$ is always ≥ 0 and equals 0 only when $p=q$ everywhere.

Interpretation 1: How independent are 2 variables?

$D_{KL}(p(x,y)||p(x)p(y))$ asks how far is the actual joint probability from the independent joint probability.

If X and Y are independent, $D_{KL}(p(x,y)||p(x)p(y)) = 0 \Rightarrow I(X; Y) = 0$

If they are highly dependent, $D_{KL}(p(x,y)||p(x)p(y))$ is large $\rightarrow I(X; Y)$ is large

Interpretation 2: Cross Entropy

True distribution = P but you don't know it, so you use Q as your best guess for encoding.

If P was known, encoding = $H(P)$ bits, the minimum. Since you are using Q instead, you pay a penalty of $D_{KL}(P||Q)$.

$$\text{So Cross entropy} = -\sum_x p(x) \log(Q(x)) = H(P) + D_{KL}(P||Q)$$

Discrete Time Markov Chains

States - represented by circles

Transition Probabilities going out of any state should add to 1. Ergodic Thm for finite, irreducible, aperiodic MC

$\pi = \Pi P$, Π is a vector of the probability distribution of X_n . $\Pi = [\pi_0 \ \pi_1 \ \dots \ \pi_n]$

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & p & 0 & 1-p \\ 0 & p/2 & p/2 & 0 \end{bmatrix}$$

All rows add to 1.

Represent final states with a self loop of $p=1$.

Solving:

① Write out balance equations and $\sum_{i=0}^n \pi_i = 1$

π_i represents fraction of time spent at state i .

$$[\pi_0 \ \pi_1 \ \pi_2] = [\pi_0 \ \pi_1 \ \pi_2] \begin{bmatrix} 1 & 0 & 0 \\ p & 0 & 1-p \\ p/2 & p/2 & 0 \end{bmatrix}$$

$$\pi_0 = \pi_0 + \pi_1 p + \pi_2 \frac{p}{2}$$

$$\pi_1 = \pi_1 \frac{p}{2}$$

$$\pi_2 = \pi_1 (1-p)$$

$$\pi_0 + \pi_1 + \pi_2 = 1$$

Solve for all π_i in terms of p .

(chain always converges to π)

Markov Chains Classifications

① Accessibility - State j is accessible from state i if there's a nonzero probability path from i to j in some number of steps.

② Communication - States i and j communicate if you can get from i to j and vice versa.

③ Irreducible - A chain is irreducible if all states communicate with each other.

④ Recurrent - $P(X_n = i | X_0 = i) = 1$. Given that state 0 is i , you are guaranteed to return back to i eventually.

Transient - $P(X_n = i | X_0 = i) < 1$. Given that state 0 is i , there is a chance you never return to i .

⑤ Aperiodic - GCD of all paths from a node i to itself is 1.

Period of State i : Find all the possible paths to go from state i back to state i . Record the step

count for each. Take the gcd of all the step

counts to get the period of State i .

IF A MARKOV CHAIN IS IRREDUCIBLE, ALL STATES

HAVE THE SAME PERIOD!!!

Aperiodic

Period of all states is 1.

Fundamental Theorem of Markov Chains

If a Markov Chain is irreducible and aperiodic, then

for any starting distribution (i.e. {0.5 0.5} means 50% chance

start in state 0 and 50% in state 1), the probability of

being in state i at step n converges to the same value

π_i from the invariant distribution π .

Formally, if a Markov Chain is irreducible and aperiodic,

then for any initial distribution π_0 , we have that

$\pi_n \rightarrow \pi$ as $n \rightarrow \infty$, and π is the unique invariant distribution

for the Markov chain.

Markov Chain Applications

Let X = # of steps before reaching state A:

$\alpha(i) = 0$ if $i = A \Rightarrow$ Already at A

$\alpha(i) = 1 + \sum_j P(i,j) \alpha(j) \Rightarrow$ Step from i to j plus future steps

Probability of Reaching A before B:

$\alpha(i) = 1$ if $i = A \Rightarrow$ Already at A

$\alpha(i) = 0$ if $i = B \Rightarrow$ Already at B

$\alpha(i) = \sum_j P(i,j) \alpha(j) \Rightarrow$ Land in state j with $p = P(i,j)$ and

future steps = $\alpha(j)$

Absorption

MC with states $S = \{0, 1, \dots, r\}$

state 1 is recurrent.

Let $\alpha(i) = P(\text{absorbed in 1} | X_0 = i)$

Then $\alpha(i) = 1$, $\alpha(j) = 0$ for all other absorbing states

$$\alpha_k = \sum_k \alpha_k P_{kk}$$

Mean Return Time: Consider the mean return time

to state $l \in S$.

$r_l = 1 + \sum_k t_k P_{lk}$ where t_k is the expected time to hit l given $X_0 = k$.

Then $t_l = 0$, $t_k = 1 + \sum_j t_j P_{kj}$

$$r_l = \frac{1}{\pi_l}$$

In general, let B_i be the quantity you care

about at the i th state. Decide which subset of

states exist such that $B = 0$ at those states.

Then $B_k = \alpha(k) + \sum_l B_l P_{kl}$. $\alpha(k)$ is the immediate cost of

getting to state k . Then for all states, l , you can transition

to from k , sum its $B_k \cdot P_{kl}$, where P_{kl} is the probability of

going from k to l .

Markovian Reversibility

Running backwards is the same as forwards.

(check: The probability of being in state i and jumping to j

equals the probability of being in j and jumping to i)

Detailed Balance: $\pi_i P(i,j) = \pi_j P(j,i)$

To guarantee π is stationary distribution:

$$\sum_i \pi(i) P(i,j) = \sum_i \pi(j) P(j,i) = \pi(j) \sum_i P(j,i) = \pi(j)$$

Ex: $\textcircled{0} \dots \textcircled{0} \xrightarrow{\lambda_{i,i+1}} \textcircled{0} \xleftarrow{\lambda_{i+1,i}} \textcircled{0} \dots \textcircled{0}$

DB: $\pi_n \lambda_n = \pi_{n+1} \lambda_{n+1}$ and $\sum \pi_i = 1 \Rightarrow \pi(n) = \pi_0 \prod_{k=0}^{n-1} \frac{\lambda_k}{\lambda_{k+1}}$ FIND THE PATTERN!

$$\pi(0) \lambda_0 = \pi(1) \lambda_1 \Rightarrow \pi(1) = \pi(0) \frac{\lambda_0}{\lambda_1} \Rightarrow \pi(2) = \pi_0 \cdot \frac{\lambda_0}{\lambda_1} \cdot \frac{\lambda_1}{\lambda_2}$$

Shannon's Random Coding Scheme

Randomly generate 2^m codewords of length n . To transmit message k , send codeword k through the channel. A random codeword matches on all non-erased bits with $p = 2^{-nC}$.

Conditional Entropy

$$H(Z|X_1) = \sum_{x_1} P(X_1 = x_1) H(Z|X_1 = x_1)$$

$$H(Z|X_1) = H(Z, X_1) - H(X_1)$$

$\phi(z) = P(Z \leq z)$ shaded area to the left of z
sometimes $Q(z)$ but this is the Gaussian CDF.

Handshake lemma: The sum of degrees of a graph is $2E$.

Detailed Balance is used to verify a proposed stationary distribution π and check reversibility.

If a stationary distribution exists for a finite and irreducible MC, all states are positive recurrent.

Positive recurrent means recurrent and $E[T_i] < \infty$. So for any state i , you will return back to that state in finite time.

$$\pi(i) = \frac{1}{E[T_i]}$$

$X_0, X_1, \dots$ = Markov Chain with transition matrix:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, S = \{0, 1, 2\}$$

$Y_i = \begin{cases} 1 & \text{if } X_i = 1 \\ 0 & \text{otherwise} \end{cases}$ Show $Y_0, Y_1, \dots$ is not a Markov chain.

Solution:

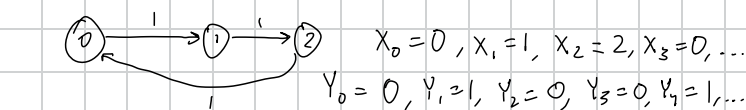
Markov Property: The past gives no extra info about the future beyond the present.

$$P(Y_{i+1} = y | Y_i) = P(Y_{i+1} = y | Y_i, X_i)$$

WTS: Markov Property fails:

Two scenarios where:

- ① Present is the same
- ② Past differs
- ③ Future differs



Let $y = 1$

$P(Y_{i+1} = 1 Y_i = 0, Y_{i-1} = 0) = 1$	Past	future
$P(Y_{i+1} = 1 Y_i = 0, Y_{i-1} = 1) = 0$	$\{0, 0, 1\}$	$\{1, 0, 0\}$
		present

The Markov property fails so $Y_0, Y_1, \dots$ is not a MC.

BEC Channel with erasure probability $p = 0.4$. Use channel $n = 100,000$ times to transmit message of length m bits.

a) Max length of m ?

Sol For BEC, $C = 1 - p = 1 - 0.4 = 0.6$

$$\max m = nC = 100000(0.6) = \boxed{60000 \text{ bits}}$$

b) Max value of m using Shannon's coding scheme if $P(\text{error}) \leq 2^{-100}$

codewords $\downarrow$ $P(\text{error for 1 codeword})$

$$P(\text{error}) = 2^m \cdot 2^{-nC} = 2^{m-nC} \leq 2^{-100}$$

$$m - nC \leq -100$$

$$m \leq nC - 100$$

$$m \leq 100000(0.6) - 100$$

$$m \leq 60000 - 100$$

$$\boxed{m \leq 59900 \text{ bits}}$$

$X_1 \sim \text{Bernoulli}(p_1), X_2 \sim \text{Bernoulli}(p_2)$, independent coin flips. $Z = X_1 + X_2$. What is $H(Z|X_1)$?

$$H(Z|X_1) = H(Z, X_1) - H(X_1)$$

Since $Z = X_1 + X_2$, knowing Z and X_1 means you know X_2 and knowing X_1 and X_2 means you know Z , so $H(Z, X_1) = H(X_1, X_2)$

$$H(Z|X_1) = H(X_1, X_2) - H(X_1)$$

$H(Z, X_1) = H(X_1) + H(X_2) - H(X_1)$ since X_1, X_2 are independent.

$$H(Z|X_1) = H(X_2)$$

$$H(Z|X_1) = p_2 \cdot \log_2\left(\frac{1}{p_2}\right) + (1-p_2) \cdot \log_2\left(\frac{1}{1-p_2}\right)$$

$\lambda > 0, n = 100, X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\lambda)$. Use CLT to find b where $P(\frac{1}{n} \sum_{i=1}^n X_i < b) = \frac{1}{10}$

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{CLT says } \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

$$E[X_i] = \lambda \quad \sigma^2 = \text{Var}(X_i) = \lambda \quad \text{so } \frac{\bar{X}_n - \lambda}{\sqrt{\lambda/n}} \xrightarrow{d} N(0, 1)$$

$$P(\bar{X}_n < b) = \frac{1}{10} \rightarrow P\left(\frac{\bar{X}_n - \lambda}{\sqrt{\lambda/n}} < \frac{b - \lambda}{\sqrt{\lambda/n}}\right) = \frac{1}{10}$$

$$\text{Large } n \text{ and } < \text{ so } \phi\left(\frac{b - \lambda}{\sqrt{\lambda/n}}\right) = \frac{1}{10}$$

$$\frac{b - \lambda}{\sqrt{\lambda/n}} = \phi^{-1}\left(\frac{1}{10}\right)$$

$$b - \lambda = \phi^{-1}\left(\frac{1}{10}\right) \cdot \frac{\sqrt{\lambda}}{\sqrt{n}}$$

$$b = \phi^{-1}\left(\frac{1}{10}\right) \cdot \frac{\sqrt{\lambda}}{\sqrt{n}} + \lambda$$

$$b = \phi^{-1}\left(\frac{1}{10}\right) \cdot \frac{\sqrt{\lambda}}{\sqrt{100}} + \lambda$$

$$b = \phi^{-1}\left(\frac{1}{10}\right) \cdot \frac{\sqrt{\lambda}}{10} + \lambda$$

Connected graph with V vertices and E edges. Consider a random walk on this graph. From any vertex v , move to one of its adjacent vertices with $p = \frac{1}{d(v)}$ where $d(v)$ = degree of v .

a) Find $\pi(v)$

$\pi(v)$ = amt of time spent at v / total time

To get to v , you have to arrive from one of $d(v)$ edges. There are $2|E|$ total edges. So $\pi(v) = \frac{d(v)}{2|E|}$

Detailed Balance: $\pi_i P(i, j) = \pi_j P(j, i)$

$$\pi(i) \cdot \frac{1}{d(i)} = \pi(j) \cdot \frac{1}{d(j)}$$

$$\frac{d(i)}{2|E|} \cdot \frac{1}{d(i)} = \frac{d(j)}{2|E|} \cdot \frac{1}{d(j)}$$

$$\frac{1}{2|E|} = \frac{1}{2|E|} \quad \checkmark$$

Detailed balance holds so $\pi(v) = \frac{d(v)}{2|E|}$.

b) Show that MC is reversible.

Detailed balance holds so the MC is reversible.

c) $P(i, j) = P(j, i)$ for all pairs of states, no self loops. Prove positive recurrent for all states.

Probably spend equal time at each vertex so

$$\pi(i) = \frac{1}{|V|}$$

$$\text{Detailed balance: } \pi(i) = \frac{1}{|V|} = \pi(j) \quad \checkmark$$

$$\text{Chain is finite so } E[T_i] = \frac{1}{\pi(i)} = |V| < \infty \quad \checkmark$$

Irreducible: The graph is connected and the condition $P(i, j) = P(j, i)$ means you can get from vertex i to vertex j for any $i, j \in V$. So all states communicate and the graph is irreducible. $\checkmark$

So the MC is finite and irreducible so it is positive recurrent.

Conditional Probability

Event Partitioning: $A = A_1 \cup A_2 \cup \dots \cup A_n$ and $A_1, \dots, A_n$ are mutually exclusive.

$$\text{Total Probability Rule: } P[B] = \sum_{i=1}^n P[B \cap A_i] = \sum_{i=1}^n P[B|A_i]P[A_i]$$

Since A_i covers all of the sample space, so you can sum the probabilities of B and A_i happening at the same time.

Independence: Events A and B are independent if knowing one happened doesn't change the probability of the other.

$$P(A|B) = P(A) \quad \text{OR}$$

$$P(B|A) = P(B) \quad \text{OR}$$

$$P(A \cap B) = P(A)P(B)$$

Pairwise Independence: Assume events A, B, C . These are pairwise independent if AB, BC , and AC are all independent

Mutual Independence

Assume events $A_1, A_2, \dots, A_n$. They are mutually independent if every possible group of them is independent. That means every pair, triple, etc. are independent. Pairwise independence doesn't imply mutual independence.

$$P[\bigcap_{i \in I} A_i] = \prod_{i \in I} P(A_i)$$

The set of events A_i is mutually independent if the probability that they all happen at the same time is equal to the product of their individual probabilities.

Mutually Exclusive Events

$$P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i) \quad \text{If events are mutually exclusive, the}$$

probability of their union is the sum of their individual probabilities as there is no double counting.

Basics

Conditional Probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$, $P(B) > 0$

Total Probability: $P(B) = \sum_{i=1}^n P(A_i) P(B|A_i)$

Bayes Rule: $P(A|B) = \frac{P(A) P(B|A)}{P(B)}$

Union Bound: $P(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} P(A_i)$

Independence: $P(A|B) = P(A)$ or $P(A \cap B) = P(A) P(B)$

Pairwise Independence: every pair of RVs is independent (does not imply mutual)

Conditional Independence: every subset of of RVs is independent (implies pairwise)

Random Variables

Discrete	PMF / PDF / CDF	$\mathbb{E}[X]$	$\text{Var}(X)$	MGF	
Uniform (a, b)	$\Pr(X=k) = \frac{1}{b-a+1}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$	$\frac{1}{b-a+1} \frac{e^{bs}-e^{as}}{1-e^s}$	each # between $[a, b]$ equally likely
Bernoulli (p)	$\Pr(X=0) = 1-p$; $\Pr(X=1) = p$	p	$p(1-p)$	$1-pe^s$	one coin flip w/ prob p
Binomial (n, p)	$\Pr(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$	$(1-pe^s)^n$	# of successes in n coin flips
Geometric (p)	$\Pr(X=k) = (1-p)^{k-1} p$	$1/p$	$\frac{1-p}{p^2}$	$\frac{pe^s}{1-(1-p)e^s}$	# of trials before first success
Poisson (λ)	$\Pr(X=k) = \frac{\lambda^k}{k!} e^{-\lambda}$	λ	λ	$e^{\lambda(e^s-1)}$	# of events in window
Exponential (λ)	$f_X(x) = \lambda e^{-\lambda x}$ $F_X(x) = 1-e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$	$\frac{\lambda}{\lambda-s}$	waiting time to next event
Gaussian (μ, σ)	$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$e^{s\mu + \frac{\sigma^2 s^2}{2}}$	shape everything converges to
Uniform (a, b)	$f_X(x) = \frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$s \neq 0? \frac{e^{bs}-e^{as}}{s(b-a)}: 0$	equally likely anywhere in range
Erlang (k, λ)	$\Pr(X=x) = (\lambda^k x^{k-1} e^{-\lambda x}) / (k-1)!$	k/λ	k/λ^2	$(\lambda / (\lambda-s))^k$	waiting time until k th event; sum of k exp

MGFs

$M_X(s) = \mathbb{E}[e^{sX}] = \sum_k e^{sk} P(X=k) = \int_{-\infty}^{\infty} e^{sx} f_X(x) dx$

Derivative of an MGF: $\left. \frac{d^n M(s)}{ds^n} \right|_{s=0} = \int x^n f(x) dx = \mathbb{E}[X^n]$

Uniqueness of MGF: $M_X(s) = M_Y(s) \Rightarrow X=Y$

$\{X_n\} = X_1, X_2, \dots, X_n$ (iid RVs)

Convergence

almost surely $\Rightarrow$ in probability $\Rightarrow$ in distribution

Sample/Empirical Mean: $M_n = \frac{1}{n} \sum X_i$

almost surely: $(X_n)_{n=1}^{\infty}$ converges a.s. to X if $P(\{ \omega \in \Omega : \lim_{n \rightarrow \infty} X_n = X \}) = 1$ or $P(\lim_{n \rightarrow \infty} X_n \neq X) = 0$ or $X_n \xrightarrow{a.s.} X$

True/Population Mean: $\mathbb{E}[M_n] = \frac{1}{n} \sum \mathbb{E}[X_i] = \mathbb{E}[X_i]$

Variance: $\text{Var}(M_n) = \frac{1}{n^2} \sum \text{Var}(X_i) = \frac{n \text{Var}(X_i)}{n^2} = \frac{\text{Var}(X_i)}{n}$
• as $n \rightarrow \infty$, $\text{var}(M_n) = 0$

SLLN: $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} \mathbb{E}(X_i)$

WLLN: $\lim_{n \rightarrow \infty} P(|\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X]| \geq \epsilon) = 0$

Borel-Cantelli: $\sum_{n=1}^{\infty} P(A_n) < \infty \Rightarrow P(A_n \text{ i.o.}) = 0$

CLT: $\frac{\bar{X}_n - \mu}{\sqrt{\frac{\sigma^2}{n}}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1)$

$\sum_{n=1}^{\infty} P(A_n) = \infty$ and $(A_n)_{n=1}^{\infty}$ independent $\Rightarrow P(A_n \text{ i.o.}) = 1$

Concentration Bounds

Markov Bound: $P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$ for non-neg X

in probability: $(X_n)_{n=1}^{\infty}$ converges i.p. to X if $\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) = 0$ or $X_n \xrightarrow{p} X$

Chebyshev: $P(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}$ for $c > 0$

Chernoff: $P(X > a) \leq \min_{s>0} \frac{M_X(s)}{e^{sa}}$ (right tail)
 $P(X < a) \leq \min_{s>0} \frac{\mathbb{E}[e^{-sX}]}{e^{-sa}}$ (right tail)

in distribution: $(X_n)_{n=1}^{\infty}$ converges i.d. to X if $\lim_{n \rightarrow \infty} P(X_n \leq x) = P(X \leq x)$ $X_n \xrightarrow{d} X$

SLLN/Borel-Cantelli/Big $\Rightarrow$ almost surely Chebyshev $\Rightarrow$ in probability CLT $\Rightarrow$ in distribution

Entropy

Information Theory

$H(X) = \mathbb{E}[\log \frac{1}{P(X)}] = \sum_x P(X=x) \log(\frac{1}{P(X=x)}) = - \sum_x P(X=x) \log P(X=x)$; average surprise/info from observing X

Conditional Entropy: $H(X|Y) = \sum_y P(Y=y) H(X|Y=y)$
 $= \sum_y P(Y=y) \sum_x P(X|Y=y) \log \frac{1}{P(X|Y=y)}$; average remaining uncertainty in X after Y
 $= H(X, Y) - H(Y)$

Chain Rule of Entropy: $H(X, Y) = H(Y) + H(X|Y) = H(X) + H(Y|X)$

Mutual Information: $I(X; Y) = H(X) - H(X|Y) = H(X) + H(Y) - H(X, Y)$
• how much information Y gives about X

Channel Capacity: $C = \max_{P_X} I(X; Y)$; mutual information maximized over input distribution, X
 $\hookrightarrow I(X; Y) = H(Y) - H(Y|X) \leftarrow$ fixed
 $\uparrow$ maximize this

Erosure Channels & Coding

Binary Erasure Channel: marks bits as erased w/ probability p
Capacity = $1-p$

Binary Symmetric Channel: flips bits w/ probability p
Capacity = $1 - H(p) = 1 - [-p \log p - (1-p) \log (1-p)]$

Shannon's random coding

message of L bits
encode each 2^L possible messages to an n bit codeword
rate $R = L/n$

BSC: prob of a wrong codeword matching on noisy bits = $\frac{1}{2^{n(1-H(p))}}$

BEC: prob of a wrong codeword matching on unerased bits = $\frac{1}{2^{n(1-p)}}$